

Министерство образования Республики Беларусь
Учреждение образования «Белорусский государственный университет
информатики и радиоэлектроники»

Факультет компьютерных систем и сетей
Кафедра Информатики
Дисциплина «Конструирование программ»

ОТЧЕТ
К лабораторной работе №1
на тему:
«СИСТЕМА КОНТРОЛЯ ВЕРСИЙ GIT»
6-05-0612-02 101

Выполнил студент группы 453504
СВИРКО Тимур Сергеевич

(дата, подпись студента)

Проверил доцент каф. Информатики
ЖВАКИНА Анна Васильевна

(дата, подпись преподавателя)

Минск 2025

ХОД ВЫПОЛНЕНИЯ РАБОТЫ

```
PS D:\453504_SVIRKO_24> git init
Initialized empty Git repository in D:/453504_SVIRKO_24/.git/
PS D:\453504_SVIRKO_24> git add .
warning: adding embedded git repository: IGI/LR1/geometric_lib
hint: You've added another git repository inside your current repository.
hint: Clones of the outer repository will not contain the contents of
hint: the embedded repository and will not know how to obtain it.
hint: If you meant to add a submodule, use:
hint:
hint:   git submodule add <url> IGI/LR1/geometric_lib
hint:
hint: If you added this path by mistake, you can remove it from the
hint: index with:
hint:
hint:   git rm --cached IGI/LR1/geometric_lib
hint:
hint: See "git help submodule" for more information.
hint: Disable this message with "git config set advice.addEmbeddedRepo false"
PS D:\453504_SVIRKO_24> git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   IGI/LR1/foo/111/111Test.txt
    new file:   IGI/LR1/foo/bar/222/222Test.txt
    new file:   IGI/LR1/foo/bar/barTest.txt
    new file:   IGI/LR1/foo/bar/baz/333/333Test.txt
    new file:   IGI/LR1/foo/bar/baz/bazTest.txt
    new file:   IGI/LR1/foo/bar/baz/bin/444/444Test.txt
    new file:   IGI/LR1/foo/bar/baz/bin/binTest.txt
    new file:   IGI/LR1/foo/bar/baz/bin/example/200/200Test.txt
    new file:   IGI/LR1/foo/bar/baz/bin/example/exampleTest.txt
    new file:   IGI/LR1/foo/bar/baz/bin/ln/700/700Test.txt
    new file:   IGI/LR1/foo/bar/baz/bin/ln/lnTest.txt
    new file:   IGI/LR1/foo/bar/baz/bin/readme.txt
    new file:   IGI/LR1/foo/bar/baz/bin/test.txt
    new file:   IGI/LR1/foo/fooTest.txt
    new file:   IGI/LR1/geometric_lib
    new file:   IGI/LR2/LR2Test.txt
    new file:   IGI/LR3/LR3Test.txt
    new file:   IGI/LR4/LR4Test.txt
    new file:   IGI/LR5/LR5Test.txt
    new file:   STRWEB/LR1/LR1Test.txt
    new file:   STRWEB/LR2/LR2Test.txt
    new file:   STRWEB/LR3/LR3Test.txt
    new file:   STRWEB/LR4/LR4Test.txt
```

Рисунок 1 – Создание репозитория

```
PS D:\453504_SVIRKO_24> git add IGI/LR1/Report1.docx
```

Рисунок 2 – Индексация файла

```
PS D:\453504_SVIRKO_24> git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   IGI/LR1/Report1.docx

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   IGI/LR1/Report1.docx

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    IGI/LR1/~$eport1.docx

PS D:\453504_SVIRKO_24> git commit -m "Modify report"
[main be43381] Modify report
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 IGI/LR1/Report1.docx
```

Рисунок 3 – Коммит измененного файла

```

D:\453504_SVIRKO_24\IGI\LR1> git log
commit b8a34961295cf935f3504caad7feae4a059aa2b4 (HEAD -> IGI-LR1)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 19:01:51 2026 +0300

    Add a file in LR1

commit be433815dfefb5ff3ac7f0e9289a78fa18d0ddd (main, STRWEB, IGI-LR5, IGI-LR4, IGI-LR3, IGI-LR2, IGI)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 18:36:00 2026 +0300

    Modify report

commit 55b8b472e2ebb510ac6dc6bd3991bf2a201a85fe
Author: kopachek <kopachekpat@gmail.com>
Date: Sat Feb 14 00:37:21 2026 +0300

    Initial commit

PS D:\453504_SVIRKO_24\IGI\LR1> git log --since="2 days ago"
commit b8a34961295cf935f3504caad7feae4a059aa2b4 (HEAD -> IGI-LR1)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 19:01:51 2026 +0300

    Add a file in LR1

commit be433815dfefb5ff3ac7f0e9289a78fa18d0ddd (main, STRWEB, IGI-LR5, IGI-LR4, IGI-LR3, IGI-LR2, IGI)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 18:36:00 2026 +0300

    Modify report

PS D:\453504_SVIRKO_24\IGI\LR1> git log --after="2026-03-01"
commit b8a34961295cf935f3504caad7feae4a059aa2b4 (HEAD -> IGI-LR1)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 19:01:51 2026 +0300

    Add a file in LR1

commit be433815dfefb5ff3ac7f0e9289a78fa18d0ddd (main, STRWEB, IGI-LR5, IGI-LR4, IGI-LR3, IGI-LR2, IGI)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 18:36:00 2026 +0300

    Modify report

PS D:\453504_SVIRKO_24\IGI\LR1> git log --author="kopachek"
commit b8a34961295cf935f3504caad7feae4a059aa2b4 (HEAD -> IGI-LR1)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 19:01:51 2026 +0300

    Add a file in LR1

commit be433815dfefb5ff3ac7f0e9289a78fa18d0ddd (main, STRWEB, IGI-LR5, IGI-LR4, IGI-LR3, IGI-LR2, IGI)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 18:36:00 2026 +0300

    Modify report

commit 55b8b472e2ebb510ac6dc6bd3991bf2a201a85fe
Author: kopachek <kopachekpat@gmail.com>
Date: Sat Feb 14 00:37:21 2026 +0300

    Initial commit

PS D:\453504_SVIRKO_24\IGI\LR1> git log
commit b8a34961295cf935f3504caad7feae4a059aa2b4 (HEAD -> IGI-LR1)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 19:01:51 2026 +0300

    Add a file in LR1

diff --git a/IGI/LR1/Report1.docx b/IGI/LR1/Report1.docx
index 522ac98..d99460b 100644
--- a/IGI/LR1/Report1.docx
+++ b/IGI/LR1/Report1.docx
@@ -0,0 +1,1 @@
+
\ No newline at end of file

```

Рисунок 4 – Флаги git log

```

PS D:\453504_SVIRKO_24\IGI\LR1> git add .\test.txt
PS D:\453504_SVIRKO_24\IGI\LR1> git commit -m "Add test"
[IGI-LR1 36d197e] Add test
1 file changed, 1 insertion(+)
create mode 100644 IGI/LR1/test.txt
PS D:\453504_SVIRKO_24\IGI\LR1> git reset --hard HEAD^
HEAD is now at b8a3496 Add a file in LR1
PS D:\453504_SVIRKO_24\IGI\LR1> git log
commit b8a34961295cf935f3504caad7feae4a059aa2b4 (HEAD -> IGI-LR1)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 19:01:51 2026 +0300

    Add a file in LR1

commit be433815dfefbf5ff3ac7f0e9289a78fa18d0ddd (main, STRWEB, IGI-LR5, IGI-LR4, IGI-LR3, IGI-LR2, IGI)
Author: kopachek <kopachekpat@gmail.com>
Date: Wed Mar 4 18:36:00 2026 +0300

    Modify report

commit 55b8b472e2ebb510ac6dc6bd3991bf2a201a85fe
Author: kopachek <kopachekpat@gmail.com>
Date: Sat Feb 14 00:37:21 2026 +0300

    Initial commit

```

Рисунок 5 – Удаление коммита

```

PS D:\453504_SVIRKO_24\IGI\LR1> git add .\afile.txt
PS D:\453504_SVIRKO_24\IGI\LR1> git restore --staged .\afile.txt
PS D:\453504_SVIRKO_24\IGI\LR1> git status
On branch IGI-LR1
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   afile.txt

no changes added to commit (use "git add" and/or "git commit -a")
PS D:\453504_SVIRKO_24\IGI\LR1> git restore .\afile.txt

```

Рисунок 6 – Отмена индексации и восстановление файла

```

PS D:\453504_SVIRKO_24> git remote add origin https://github.com/kopachek/IGI-LR1.git
PS D:\453504_SVIRKO_24> git remote -v
origin https://github.com/kopachek/IGI-LR1.git (fetch)
origin https://github.com/kopachek/IGI-LR1.git (push)
PS D:\453504_SVIRKO_24> git push origin IGI-LR1
Enumerating objects: 49, done.
Counting objects: 100% (49/49), done.
Delta compression using up to 16 threads
Compressing objects: 100% (33/33), done.
Writing objects: 100% (49/49), 704.08 KiB | 11.18 MiB/s, done.
Total 49 (delta 8), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), done.
To https://github.com/kopachek/IGI-LR1.git
 * [new branch]      IGI-LR1 -> IGI-LR1
PS D:\453504_SVIRKO_24> git commit -m "Remove report file"
[IGI-LR1 08f3f71] Remove report file
2 files changed, 0 insertions(+), 0 deletions(-)
delete mode 100644 IGI/LR1/Report1.docx
delete mode 100644 IGI/LR1/~WRL0532.tmp
PS D:\453504_SVIRKO_24> git push origin IGI-LR1
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 361 bytes | 361.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/kopachek/IGI-LR1.git
717bdf3..08f3f71 IGI-LR1 -> IGI-LR1

```

```

PS D:\453504_SVIRKO_24> git checkout IGI-LR1
Switched to branch 'IGI-LR1'
PS D:\453504_SVIRKO_24> git add .
PS D:\453504_SVIRKO_24> git status
On branch IGI-LR1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   IGI/LR1/toTestBranch1.txt
        new file:   IGI/LR1/toTestBranch2.txt

PS D:\453504_SVIRKO_24> git commit -m "Add files to move to branch"
[IGI-LR1 f030b3a] Add files to move to branch
 2 files changed, 2 insertions(+)
   create mode 100644 IGI/LR1/toTestBranch1.txt
   create mode 100644 IGI/LR1/toTestBranch2.txt
PS D:\453504_SVIRKO_24> git checkout Test_branch_1
Switched to branch 'Test_branch_1'
Your branch is up to date with 'new-origin/Test_branch_1'.
PS D:\453504_SVIRKO_24> git checkout IGI-LR1 -- IGI/LR1/toTestBranch1.txt IGI/LR1/toTestBranch2.txt
PS D:\453504_SVIRKO_24> git commit -m "Add test files"
[Test_branch_1 d9b387b] Add test files
 2 files changed, 2 insertions(+)
   create mode 100644 IGI/LR1/toTestBranch1.txt
   create mode 100644 IGI/LR1/toTestBranch2.txt
PS D:\453504_SVIRKO_24> git rm .\IGI\LR1\toTestBranch2.txt
rm 'IGI/LR1/toTestBranch2.txt'

```

Рисунок 7 – Копирование и удаление файла

```

PS D:\453504_SVIRKO_24> git commit -m "Add file to merge"
[Test_branch_2 140d407] Add file to merge
 1 file changed, 1 insertion(+)
   create mode 100644 IGI/LR1/toMerge.txt
PS D:\453504_SVIRKO_24> git checkout IGI-LR1
Switched to branch 'IGI-LR1'
PS D:\453504_SVIRKO_24> git merge Test_branch_2
Updating f030b3a..140d407
Fast-forward
   IGI/LR1/toMerge.txt | 1 +
   1 file changed, 1 insertion(+)
   create mode 100644 IGI/LR1/toMerge.txt
PS D:\453504_SVIRKO_24> git push new-origin --delete Test_branch_2
To https://github.com/kopachek/IGI-LR1.git
 - [deleted]      Test_branch_2
PS D:\453504_SVIRKO_24> git push new-origin IGI-LR1
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 16 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (11/11), 880 bytes | 440.00 KiB/s, done.
Total 11 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/kopachek/IGI-LR1.git
   08f3f71..140d407  IGI-LR1 -> IGI-LR1

```

Рисунок 8 – Слияние и удаление ветки